THE NORMALISED DUALIZING COMPLEX IS NORMALISED AT CLOSED POINTS LYING OVER CLOSED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AWL (sources/stacks-duality/pinned-original-source.tex line 5805).

Allowed standard prerequisites (editorial).

- The upper shriek of a closed immersion and its compatibility with composition. Same chapter, Lemma labelled lemma-shriek-closed-immersion, cited at proof lines 5826-5830. Identifies the localised Hom with i_x^! of the dualizing complex.
- Finiteness of the residue field extension at a closed point of a finite type morphism. Morphisms, Lemma labelled morphisms-lemma-closed-point-fibre-locally-finite-type, cited at proof lines 5836-5840. Shows pi is finite, so that the composition of shrieks applies.
- The comparison between pushforward and internal Hom for a Noetherian morphism. Same chapter, Example labelled example-iso-on-RSheafHom-noetherian, cited at proof lines 5850-5852. Gives the first equality computing Rpi_* of the shriek.
- Normalised dualizing complexes and their description through the residue field. Same chapter, the definition of a normalised dualizing complex, invoked in the statement and used at proof lines 5842-5846. Turns normalisation into i_s^!omega_S = kappa(s)[0].

Source reference. Original lemma 0AA2: sources/stacks-duality/pinned-original-source.tex, source lines 4624–4624. Same-chapter pointer into the author environment lemma at source line 4624; invoked at proof line 5825 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01TF: sources/stacks-chow/tags.txt, source lines 2372–2372. Cross-chapter standard prerequisite located in the project tag list

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staged in the local snapshot; invoked at proof line 5834 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0GEW: sources/stacks-duality/pinned-original-source.tex, source lines 865–865. Same-chapter pointer into the author environment example at source line 865; invoked at proof line 5849 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a Noetherian scheme and let $\omega_S^\bullet$ be a dualizing complex. Let X be a scheme of finite type over S and let $\omega_X^\bullet$ be the dualizing complex normalized relative to $\omega_S^\bullet$. If $x \in X$ is a closed point lying over a closed point s of S, then $\omega_{X,x}^\bullet$ is a normalized dualizing complex over $\mathcal{O}_{X,x}$ provided that $\omega_{S,s}^\bullet$ is a normalized dualizing complex over $\mathcal{O}_{S,s}$.*

Proof. We may replace X by an affine neighbourhood of x, hence we may and do assume that $f : X \to S$ is separated. Then $\omega_X^\bullet = f^!\omega_S^\bullet$. We have to show that $R\mathrm{Hom}_{\mathcal{O}_{X,x}}(\kappa(x), \omega_{X,x}^\bullet)$ is sitting in degree 0. Let $i_x : x \to X$ denote the inclusion morphism which is a closed immersion as x is a closed point. Hence $R\mathrm{Hom}_{\mathcal{O}_{X,x}}(\kappa(x), \omega_{X,x}^\bullet)$ represents $i_x^!\omega_X^\bullet$ by Lemma 0AA2. Consider the commutative diagram

$$\begin{array}{ccc} x & \xrightarrow{i_x} & X \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle f} \\ s & \xrightarrow{i_s} & S \end{array}$$

By Morphisms, Lemma 01TF the extension $\kappa(x)/\kappa(s)$ is finite and hence π is a finite morphism. We conclude that

$$i_x^!\omega_X^\bullet = i_x^! f^!\omega_S^\bullet = \pi^! i_s^!\omega_S^\bullet$$

Thus if $\omega_{S,s}^\bullet$ is a normalized dualizing complex over $\mathcal{O}_{S,s}$, then $i_s^!\omega_S^\bullet = \kappa(s)[0]$ by the same reasoning as above. We have

$$R\pi_*(\pi^!(\kappa(s)[0])) = R\mathcal{H}om_{\mathcal{O}_s}(R\pi_*(\kappa(x)[0]), \kappa(s)[0]) = \widetilde{\mathrm{Hom}_{\kappa(s)}(\kappa(x), \kappa(s))}$$

The first equality by Example 0GEW applied with $L = \kappa(x)[0]$. The second equality holds because π_* is exact. Thus $\pi^!(\kappa(s)[0])$ is supported in degree 0 and we win. $\square$